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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	ODDS
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4(4.4) & SDG 8(8.5)
Project / Solution Name	FuturePath AI
Name (Leader)	Nur Faqiah Binti Mohd Noh
Email & Phone (Leader)	nurfqeh23@gmail.com 01127639342
University Affiliation(s)	Universiti Teknologi Mara (UiTM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Payroll and timekeeping clerks do largely routine, rule-based work: recording hours, computing pay and deductions and entering data into payroll systems. This is the kind of work that software tends to automate. The World Economic Forum's Future of Jobs Report 2025 places accounting, bookkeeping and payroll clerks among the roles expected to suffer the greatest reduction in employment by 2030, and the same group appears among declining roles in all four editions from 2016 to 2025. The report also appears among estimates that 39% of existing skill sets will be transformed or become outdated in that period.

Malaysia shows the same gap between AI use and AI training. The Domain 1 brief, citing the Hays Asia Salary Guide 2025, reports that 56% of professionals already use AI tools at work while only 26% have received formal training, and that MOSTI expects more than 600,000 workers to need reskilling within three to five years.

A payroll clerk facing this has no clear way to see which of their skills still matter, which roles they could realistically move into, or what to learn first. Career resources are mostly static lists that date quickly, and a one-time skills test cannot keep pace with AI progress.

This is a skills-for-decent-work problem (SDG 4, target 4.4) and an employment problem (SDG 8, target 8.5).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FuturePath AI is a web-based career transition platform designed to help Payroll and Timekeeping Clerks adapt to workforce changes driven by AI and automation. Following a human-centered approach, the system supports career decision-making without making decisions on behalf of users.

Users provide their current role and skills through a simple profile of intake. FuturePath AI then analyses workforce trends, identifies rising and declining skills, matches users with future-ready occupations using O*NET occupational data, and highlights skill gaps between their current and target roles. The system also generates personalized reskilling pathways using verified learning resources and maps each recommended role to relevant United Nations Sustainable Development Goals (SDGs).

To address uncertainty in workforce forecasting, FuturePath AI provides multiple AI-adoption scenarios (Moderate and Fast Adoption), allowing users to explore different future outcomes. Recommendations are generated using a transparent weighted matching model that combines skill similarity, workforce trend signals, and SDG alignment. All recommendations are traceable to verified sources, ensuring explainability, reliability, and responsible AI use.

Trend data is read at runtime from a dated file, with the source and date shown for every item, so updating the file changes the output. FuturePath AI never invents roles, courses or statistics: everything comes from cited data files. Workshop skills used: cloud deployment on Replit; no database or GenAI API is used in the prototype, as data is stored in CSV files read by a Flask backend.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend and deployment: Replit is used to build and run the web prototype.

Backend: Python with Flask

Data storage: CSV files stored in the project's data/ folder and read by the Flask backend. No database is used in the current prototype.

Data layer:

- Roles.csv (occupational skills)
- Trends.csv (rising and declining skills)
- Courses.csv (learning resources)
- Sdg_map.csv (SDG alignment mapping)

Development environment: VS Code and Anaconda.

Architecture: User --> Replit-hosted frontend --> Python/Flask backend --> CSV data files --> results displayed to the user.

Sponsor-provided services: Yes. Replit, the official SDC cloud platform, is used to build, run, and host the prototype. No sponsor-provided GenAI API is used. AI assistants (ChatGPT, Claude, Microsoft Copilot, Replit AI Assistant, Google Gemini) helped develop the project and are not called by the app at runtime; see the Technical Tools Declaration.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathize: we defined one user, a payroll and timekeeping clerk whose routine tasks are being automated and studied typical tasks and skills in O*NET and the WEF Future of Jobs Report 2025.

Define: we wrote one problem statement and locked scope: the six required features are in; a chatbot, resume builder and job portal are out.

Ideate: we mapped the journey from profile to pathway in four screens before choosing tools.

Prototype: we sketched the four screens on paper, then built them on Replit, keeping data in separate CSV files so the team works in parallel and the data stays updatable.

Test: we run synthetic sample profiles through the app and check that every recommended role, skill and course traces to a cited source, and that changing the trend file changes the ranking.

Deployment: The prototype is deployed via Replit and supported by a public GitHub repository under the Apache 2.0 license, ensuring transparency, accessibility, and reproducibility for judges and future development.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. SDG 8, Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Estimated manual preparation time: 25 minutes per profile to prepare and verify a basic career-transition pathway comprising a role match, skill-gap breakdown, at least one relevant course with a link and duration, and an SDG rationale. This is an initial team planning assumption, not a measured result. It will be validated through timed testing using five synthetic profiles before the Phase 2 evaluation
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Average time required to prepare and verify a complete career-transition pathway, measured in minutes per profile.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target a 20% reduction in average pathway preparation and verification time compared with the validated manual baseline, using the same five synthetic profiles and completeness criteria.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✓ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET Database	Dataset	Creative Commons (CC BY 4.0) license
WEF Future of Jobs Report 2025	Report (cited reference)	All rights reserved; cited only
UN Sustainable Development Goals (SDGs)	Framework	UN website terms; mapping written by our team
Python	Code (language)	PSF License
Flask	Code (framework)	BSD-3-Claude
Replit	Development & Deployment Platform	Proprietary
GitHub	Version Control	Proprietary
Microsoft Copilot,Google Gemini,Claude,ChatGPT,Replit	AI Development Assistants	Proprietary; see Technical Tools Declaration

6.3 Employment-Related IP Conflicts

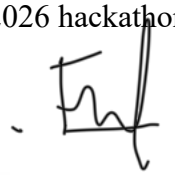
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Nur Faqiah Binti Mohd Noh]

[20.10.2026]